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ORIGINAL PAPER

# Automated multi-temporal LiDAR analysis of surface-elevation changes along pipeline corridors using ArcGIS Pro and Python

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## Abstract

Multi-temporal airborne LiDAR provides high-resolution measurements for detecting surface-elevation changes and supporting infrastructure monitoring. This study develops and evaluates workflows for measuring surface-elevation changes near crude oil pipelines in Jefferson County, Southeast Texas, using LiDAR datasets acquired in 2006 and 2017. Two approaches were compared: a conventional ArcGIS Pro-based workflow and a Python-enhanced workflow designed to automate point-cloud processing, spatial matching, and elevation-change calculations. The results revealed spatially variable patterns of surface elevation loss and gain along pipeline corridors. For a representative single LiDAR tile processed using both methods, the Python-enhanced workflow reduced processing time by approximately **77%**, while producing broadly comparable spatial patterns of elevation change despite localized differences in point-level estimates. LiDAR-derived changes were also qualitatively compared with available RTK/GNSS benchmark and InSAR observations. Because these datasets differ in acquisition periods, spatial resolution, measurement characteristics, and observation locations, they were used as independent cross-checks rather than direct validation of LiDAR-derived rates. Observed elevation losses were not interpreted as confirmed subsidence because land development, erosion, deposition, and other localized surface modifications may also contribute to the measured changes. The findings demonstrate the potential of automated multi-temporal LiDAR processing as a scalable screening approach for identifying significant surface-elevation changes near pipeline infrastructure. The Python-enhanced workflow substantially reduces processing time and manual intervention, supporting efficient analysis of larger LiDAR datasets and helping prioritize locations for further field investigation and complementary ground-deformation monitoring.

*Clinical Trial Number*: Not Applicable.

**Keywords** Point-cloud processing · Geomorphological change · Infrastructure monitoring · Remote sensing · Ground deformation · Geospatial automation

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# 1 Introduction

## 1.1 Background

LiDAR (Light Detection and Ranging) is a remote-sensing technology that determines distance by measuring the travel time of emitted laser pulses reflected from a target (Lefsky et al. 2002). Since its early development in the 1960s (Macomber 1963), LiDAR has evolved into an important source of high-resolution three-dimensional spatial data. Modern systems generate dense point clouds containing horizontal coordinates and elevation measurements that enable detailed characterization of terrain and surface features (Ryan 2013).

Advances in sensing, data storage, and computing have expanded LiDAR applications across terrain mapping (Agran et al. 2021), soil characterization (Kolker et al. 2011; Farhan et al. 2024), forest management (Sankey et al. 2017; Lefsky et al. 2002; Favarin et al. 2024), 3D urban modeling (Liu 2008), and infrastructure inspection (Lee and Gharaibeh 2022). LiDAR has also been increasingly applied to the measurement of surface deformation and elevation change, including investigations in mining environments (Dong et al. 2022) and coastal regions (Zhong et al. 2022). Its high spatial resolution and point density make it particularly useful where localized changes in ground-surface elevation are important for infrastructure and environmental monitoring (Ibironke et al., 2026). More broadly, remote sensing is increasingly being used as a screening tool for surface conditions. For example Atesoglu et al. (2025); Cetin et al. (2024a, b, 2025a, b) applied remote sensing techniques to monitor terrain changes in Turkey, including the urban heat island effects, and hydrocarbon micro-seepages; and Kucukpehlivan et al. (2023) employed GIS-based hotspot analysis to examine the impacts of urban planning on urban land areas.

The availability of repeated LiDAR surveys has further enabled multi-temporal point-cloud analysis, in which measurements acquired at different times are spatially compared to identify changes in surface geometry. Such approaches have been applied to geomorphological change detection and infrastructure deformation monitoring (Burns et al. 2010; Anders et al. 2013; Alexiou et al. 2021; Kamp et al. 2023; Kalenjuk and Lienhart 2024). In urban environments, for example, Buján et al. (2020) developed an automated approach for extracting ground observations from airborne LiDAR in the presence of buildings, vegetation, and other non-ground objects, while Dong et al. (2022) applied multi-period LiDAR alignment methods to investigate surface deformation. These studies illustrate the potential of repeated laser-scanning observations for identifying spatial patterns of surface change over time.

However, differences between multi-temporal LiDAR datasets cannot automatically be attributed to ground subsidence. The reliability of elevation-change estimates can be affected by differences in acquisition geometry, point density, ground classification, spatial registration, terrain characteristics, and processing procedures (Dong et al. 2022; Zhong et al. 2022; Kamp et al. 2023). Filtering methods such as the Iterative Robust Interpolation approach have therefore been developed to improve ground-surface extraction and reduce errors associated with complex terrain and non-ground features (Sandra et al. 2020). More fundamentally, measured elevation loss between acquisition periods may reflect several processes, including subsidence, erosion, deposition, land development, excavation, or other surface modifications. Multi-temporal LiDAR change detection should therefore be inter-

preted as a measurement of surface-elevation change, with additional evidence required to establish its underlying cause.

LiDAR consequently forms one component of a broader set of techniques for monitoring ground deformation. InSAR (Interferometric Synthetic Aperture Radar), GNSS, conventional surveys, and LiDAR provide observations with different spatial coverage, temporal resolution, and measurement characteristics, making them potentially complementary rather than directly interchangeable (Yin et al. 2025). Buján et al. (2020), for example, integrated InSAR, airborne LiDAR, GPS, and land-cover information in subsidence mapping, while de Koning et al. (2020) used both InSAR and LiDAR to examine deformation in the Netherlands as part of a dike-strengthening project. The latter study reported greater variance in LiDAR measurements than in InSAR observations, illustrating that differences between sensing methods must be considered when comparing deformation estimates. Accordingly, comparisons among LiDAR, InSAR, and ground-based observations can provide useful independent context, but quantitative validation requires adequate spatial and temporal correspondence among the datasets.

The need for spatially detailed ground-deformation information is particularly relevant to infrastructure in the U.S. Gulf Coast. Recent satellite-based analysis of vertical land motion (VLM) across the Greater Houston–Galveston region demonstrated substantial spatial variability in subsidence and highlighted the importance of incorporating high-resolution VLM information into infrastructure monitoring and hazard assessment (Buzzanga et al. 2025). Using Sentinel-1-derived VLM observations from 2016 to 2023, Buzzanga et al. (2025) reported a median subsidence rate of approximately 5.3 mm/year in Houston and showed that rates varied considerably across relatively short spatial distances. The study further demonstrated how remotely sensed VLM information can be combined with infrastructure and flood-hazard data to support identification of potentially vulnerable petrochemical facilities and more targeted management decisions.

This regional evidence is particularly important because the Gulf Coast contains extensive petrochemical and transportation infrastructure that may be affected by changes in ground elevation. Buzzanga et al. (2025) showed that VLM can contribute to changing flood exposure around critical infrastructure and emphasized the value of spatially distributed remote-sensing observations for infrastructure planning. For pipeline corridors, identifying where substantial surface-elevation changes have occurred can similarly provide an initial screening mechanism for prioritizing locations for further investigation. Such changes, however, should not by themselves be interpreted as confirmed subsidence or infrastructure risk because determining their cause and engineering significance requires additional temporally and spatially coincident observations.

Despite the growing availability of multi-temporal LiDAR, processing large point-cloud datasets remains computationally demanding and can require multiple sequential operations for ground-point extraction, format conversion, spatial matching, and elevation-difference calculation. Conventional GIS workflows can become increasingly time-consuming as the spatial extent and number of LiDAR tiles increase. This creates a need for reproducible and scalable processing approaches capable of efficiently extracting and matching multi-temporal ground observations while maintaining transparency in how surface-elevation changes are calculated.

### 1.2 Motivation and objectives

Southeast Texas contains a dense network of pipelines supporting major refineries, petrochemical facilities, and other components of the regional energy infrastructure. Many of these pipelines have operated for decades, making continued monitoring of surrounding ground conditions important for infrastructure management (Texas Almanac 2021). The region is also susceptible to land subsidence associated with groundwater and hydrocarbon withdrawal, sediment compaction, and drainage of organic soils. Removal of subsurface fluids can cause permanent compaction and corresponding surface subsidence (Poland and Davis 1969), while oxidation of drained organic soils can contribute additional elevation loss (Kolker et al. 2011; Farhan et al. 2024). Where substantial ground deformation occurs near pipelines, associated differential movement and strain may affect infrastructure integrity. Consequently, identifying locations where significant surface-elevation changes have occurred can provide useful information for prioritizing further investigation.

LiDAR is well suited to such large-area screening because it provides high-resolution three-dimensional terrain measurements over extensive areas. Compared with conventional point-based surveying, airborne LiDAR can provide much greater spatial coverage and point density while reducing the need for personnel to access difficult or hazardous locations (Yen et al. 2011; Khanal et al. 2020; Farhan et al. 2024). More importantly for this study, the availability of LiDAR observations from different acquisition periods provides an opportunity to identify spatial patterns of surface-elevation change along infrastructure corridors. However, as discussed in Sect. 1.1, these changes cannot independently establish subsidence because other geomorphological and anthropogenic processes may also alter the measured ground surface. Accordingly, this study develops and evaluates workflows for detecting surface-elevation changes that may indicate potential subsidence using airborne LiDAR datasets acquired in 2006 and 2017. Rather than establishing definitive ground-truth estimates of regional subsidence, the study evaluates the ability to efficiently extract, spatially match, and compare multi-temporal ground-elevation observations along pipeline corridors. A conventional ArcGIS Pro-based workflow is compared with a Python-enhanced workflow in terms of processing efficiency, reproducibility, and consistency of the resulting spatial patterns. Two objectives are addressed:

- Develop and compare ArcGIS Pro- and Python-enhanced workflows for extracting LiDAR-derived point features, spatially matching multi-temporal observations, and quantifying surface-elevation changes that may indicate potential subsidence along pipeline corridors in Southeast Texas.
- Qualitatively compare the LiDAR-derived elevation-change results with benchmark survey and InSAR observations to assess the consistency of the identified elevation-change patterns.

For both workflows, only ground-classified LiDAR returns were retained. For the first objective, surface-elevation change was calculated for spatially matched point pairs as the 2006 elevation minus the corresponding 2017 elevation. Therefore, positive values indicate surface-elevation loss and negative values indicate elevation gain. The Python-enhanced

workflow used ArcPy for geospatial processing, laspy for LAS/LAZ point-cloud management, Open3D and PCLPy for point-cloud processing, and NumPy and Pandas for numerical and tabular operations, together with standard Python libraries for file handling and workflow automation. All scripts developed for the study have been made publicly available through a GitHub repository to support reproducibility. Detailed descriptions of the two workflows are provided in Sect. 3.

For the second objective, please note that available GNSS benchmark and InSAR measurements were used to provide independent qualitative context for the LiDAR-derived elevation changes. The 2011 Southeast Texas benchmark survey included measurements from 340 National Geodetic Survey benchmarks and reported vertical displacement rates within the region (Kruger 2019), while a related InSAR study using Sentinel-1 observations from 2016 to 2022 reported annual surface-displacement rates generally within ±2–3 cm/year along pipelines in Southeast Texas (Shrestha 2024). Because these datasets differ from the LiDAR observations in acquisition period, spatial coverage, and measurement characteristics, they were not used as direct ground-truth validation. Instead, they provide qualitative reference information for assessing whether the direction and approximate magnitude of LiDAR-derived changes are broadly consistent with independently observed ground-deformation patterns.

The remainder of this paper is organized as follows. Section 2 describes the study area and multi-temporal LiDAR datasets. Section 3 presents the ArcGIS Pro-based and Python-enhanced processing workflows for generating point features, spatially matching observations, and calculating elevation changes. Section 4 reports the resulting surface-elevation changes, processing-time comparison, and comparisons with benchmark and InSAR observations. Section 5 discusses the spatial patterns, workflow performance, measurement discrepancies, limitations, and practical implications, and Sect. 6 concludes the whole article.

## 2 Data

This study focuses on a dense pipeline region in Jefferson County, Southeast Texas (29.9423838°N to 29.8747945°N, 93.8769752°W to 94.3135457°W), spanning approximately 67,388 acres with numerous refineries and oil and gas facilities (Fig. 1).

As mentioned before, two methods were used in this paper. The first method is to directly use ArcGIS Pro's geoprocessing tools; and the second method is to develop a set of *Python-based* workflows. To compare computational efficiency, a smaller sample along West Port Arthur Road was selected for the ArcGIS Pro method; while a much larger coverage was employed for the Python-based method (see the details in Table 1).

Two LiDAR datasets were used for both methods (Fig. 1). The first dataset, collected between July 4th – 6th, 2006, by Sanborn Mapping Company for NOAA, TWDB, and FEMA, covers Jefferson County (OCM Partners 2022). The second dataset, captured between February 22nd – March 23rd, 2017, by Sanborn for the Texas Strategic Mapping (StratMap) project, covers Jefferson, Liberty, and Chambers Counties (TNRIS 2022). Together, these datasets comprise 276 index files, with more details provided in Table 1.

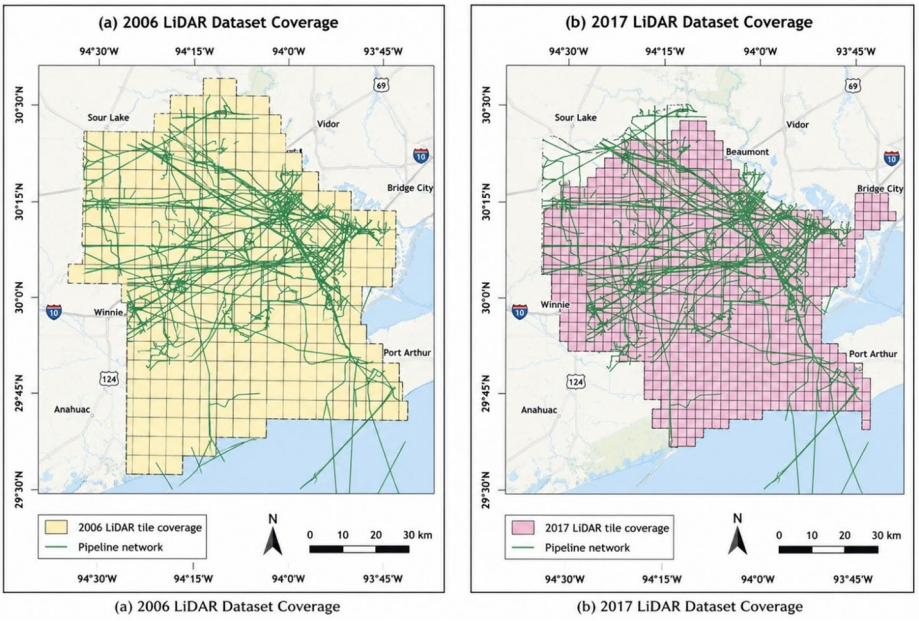



**Fig. 1** Spatial coverage of the (**a**) 2006 and (**b**) 2017 LiDAR datasets and pipeline network in Jefferson County, Texas. (Note: Green lines represent the pipeline network)

**Table 1** General attributes of the raw datasets (all at 95% confidence level)

| ArcGIS pro sample (for Method 1) | Resolution (cm) | Vertical accuracy (cm) | Horizontal accuracy (cm) | Coverage Area ($km^2$) | File size (GB) | # Points |
|---|---|---|---|---|---|---|
| 2006 dataset | 100 | 18.5 | 100 | 3.5 | 0.3 | 9,281,940 |
| 2017 dataset | 50 | 10 | 24.10 | 3.5 | 0.5 | 19,701,081 |
| Python library sample (for Method 2) | | | | | | |
| East Jefferson 2006 | 100 | 18.5 | 100 | 272.71 | 16.5 | 3,816,013,33 |
| East Jefferson 2017 | 50 | 10 | 24.10 | 272.71 | 809 | 16,413,051,679 |
| West Jefferson 2006 | 100 | 18.5 | 100 | 171.32 | 4.7 | 109,027,702 |
| West Jefferson 2017 | 50 | 10 | 24.10 | 171.32 | 71 | 1,438,970,126 |

## 3 Method

### 3.1 General method

Two methods were employed to analyze two sets of LiDAR data (one collected in 2006 and another in 2017) and extract numerical values for assessing elevation changes from 2006 to 2017 in the area along a pipeline network. Initially, the standard ArcGIS geoprocessing approach was used to convert LiDAR data to shapefiles (Environmental System Research Institute (Esri) 1990). However, the extensive processing time required for large LAZ/LAS datasets posed a challenge. To enhance efficiency, a Python-based method was developed,

significantly reducing processing time while maintaining accuracy. This improvement provides a more practical and scalable solution for large-scale LiDAR data analysis under time constraints. The workflow was intentionally developed using open-source Python libraries in conjunction with ArcGIS Pro to improve transparency, reproducibility, and adaptability for future LiDAR-based infrastructure monitoring studies.

*Method 1* automates the use of spatial analysis tools in ArcGIS Pro, converting LAS files to shapefiles through aggregated multipoint point cloud data (PCD), and XYZ formats, and then spatially joining intersecting points within the average point spacing distance to calculate elevation changes over the 11-year period. Figure 2 shows the model built in ArcGIS Pro. The detailed procedure is given Sect. 3.2

In contrast, *Method 2* introduces a Python-based workflow consisting of two steps: (1) converting LiDAR data from LAZ format to the shapefiles using Python libraries, and (2) merging points from different datasets to compute elevation changes

The primary distinction between these two methods lies in the approach used for shapefile creation: Method 1 relies entirely on ArcGIS geoprocessing tools for shapefile creation and elevation difference calculations; while on the other hand, Method 2 employs Python libraries to improve computational efficiency. A comparative analysis of these methods is shown below in Sect. 3.2 and 3.3, respectively. The consistency of the two approaches was further assessed by comparing their LiDAR-derived elevation-change patterns with independent RTK/GNSS SubSETX 2011/2023 benchmark surveys and InSAR observations from a related subsidence study. Figure 3 illustrates the methodological workflows of the Python-based method in details.

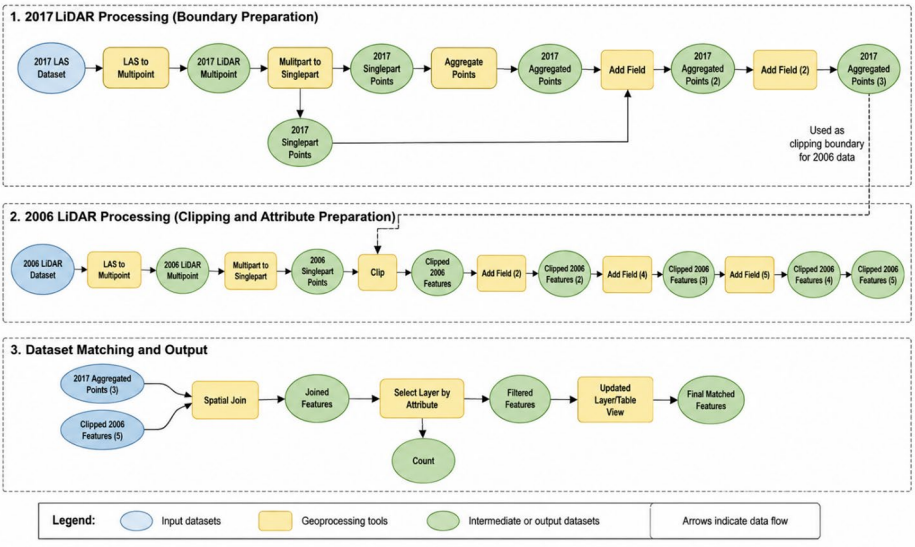



**Fig. 2** ArcGIS Pro based workflow for processing and spatially matching the 2006 and 2017 LiDAR-derived point features. (Note: Blue=input, Yellow= tools, Green= outputs)

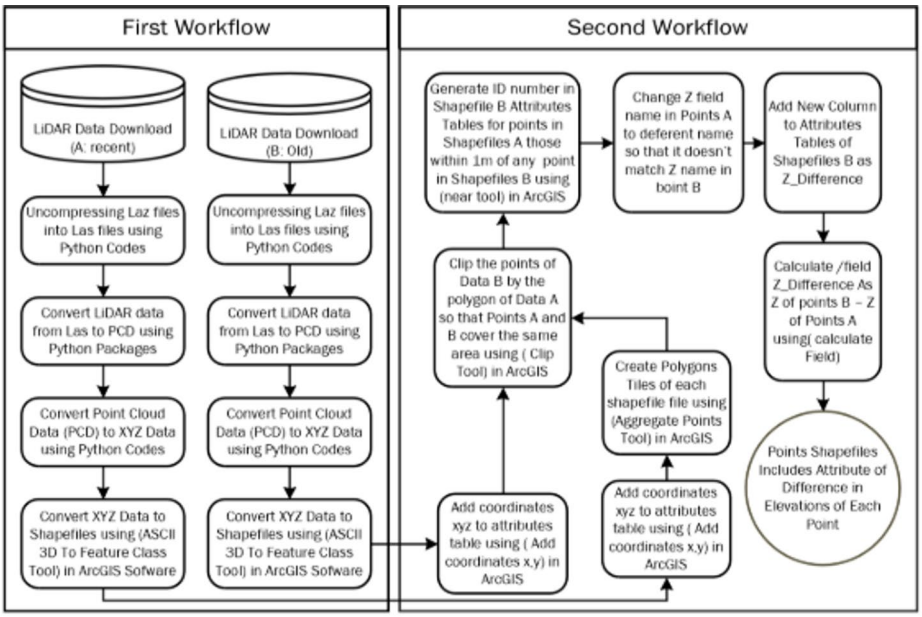



**Fig. 3** Python-based workflows for LiDAR processing and elevation change analysis

### 3.2 From LiDAR data to shapefiles using ArcGIS Pro tools (Method 1)

Based on the functions of ArcGIS Pro, as shown in Fig. 2, the workflow consists of the following eight steps designed to (1) convert LAZ/LAS files into shapefiles, (2) spatially join these shapefiles based on intersecting points, and (3) calculate the elevation differences between the intersecting points.

*Step 1*: The two LiDAR datasets in LAS format were converted to multipoint features. The LAS format supports the storage of numerous predefined attributes for each LiDAR point (ASPRS 2019). Here the Point Classification Code 2 was used to filter ground points

*Step 2*: Single-part features of both datasets were generated for each point by adding a new field, "ORIG_FID," containing unique IDs, while preserving the input features in the output feature class. Each part of a multipart feature was separated into individual features in the output class, but single-part features remained unchanged

*Step 3*: The single-part point features of the 2017 dataset were aggregated into polygon features with a count of points. This enabled analysis and visualization before being clipped with the single-part features of the 2006 dataset

*Step 4*: The tool of Clip was used to extract features of the 2006 dataset that overlay the 2017 dataset features, creating a new dataset, also referred to as a study area or area of interest (AOI). This new dataset contains a geographic subset of the features of one dataset using the boundary of the other dataset

*Step 5*: Three new fields X, Y and Z were added to the newly created feature class and the single-part features from the 2017 dataset to create fields for the coordinates and

elevation of each point in both datasets. Calculating each point's geometry attributes helped determine its spatial or geometric characteristics and location

*Step 6*: Attributes from the two features from Step 5 were joined together using the join operation based on intersecting and matching points within 1 m, the average point spacing of the 2006 dataset. This way all attributes of the join features (the 2017 dataset) were appended to attributes of the target features (the 2006 dataset) and copied to the output feature class. Two new fields, Join_Count and TARGET_FID, were added to the output feature class. The Join_Count field indicates the number of join features that match each target feature (TARGET_FID)

*Step 7*: Based on SQL expression which was written to select intersecting points from Step 6, this step modified the spatially joined points by selecting only intersecting points

*Step 8*: Elevation differences between the datasets were determined by subtracting Z-values of the 2006-point features from those of the 2017-point features

It is noteworthy that the use of a smaller sample size for the area west of Port Arthur allowed for more efficient resource management. However, this may limit the generalizability of the results and reduce the accuracy of elevation change predictions over larger areas. Smaller sample sizes can lead to potential biases or miss important variations in the dataset, especially in regions with heterogeneous characteristics. As such, while the reduced sample size improves computational efficiency, it may not capture the full spatial variability necessary for more comprehensive analyses, highlighting the trade-off between efficiency and accuracy in geospatial workflows.

### 3.3 From LiDAR data to shapefiles using Python library (Method 2)

Though the ArcGIS Pro geoprocessing tools effectively convert LiDAR data to shapefiles and calculates elevation differences, its processing time is significantly long, especially when handling large datasets. To address this challenge, an alternative Python-based workflow was developed, streamlining the conversion process while preserving accuracy. This method leverages Python libraries to automate key steps, reducing computational overhead and enhancing efficiency. The following section details the Python-enhanced approach, outlining its steps and advantages over the ArcGIS Pro method. As shown in Fig. 3, the first workflow aims to create shapefiles from LiDAR dataset through four steps.

*Step 1*: The compressed LiDAR dataset in LAZ format is extracted into an uncompressed LAS format using the Laspy library, which allows for efficient reading, filtering (specifically using Point Classification Code 2 for ground points), and writing of LAS files. This step streamlines data extraction compared to ArcGIS Pro's built-in tools, which require multiple manual operations

*Step 2*: The LAS files are converted into PCD (Point Cloud Data) format using the Pclpy library, which is optimized for handling and processing large LiDAR datasets. Unlike ArcGIS Pro, which can be resource-intensive and slower when managing large point clouds, Pclpy enables rapid format conversion while maintaining data integrity

*Step 3*: The PCD files are further transformed into XYZ format, which stores 3D spatial coordinates (X, Y, Z). This conversion was performed using the Open3D library, a specialized tool for 3D data processing. An additional advantage of this approach is the

seamless integration of datum transformations using NOAA's Vdatum tool. Note that these two datasets have different horizontal and vertical datums, respectively. The 2006 dataset undergoes a horizontal transformation from NAD83 (HARN) to NAD83 (2011) Projected UTM coordinates, while the 2006 and 2017 datasets are both aligned to NAVD 88 with Geoid 18 for vertical datum consistency. These transformations, which require multiple steps in ArcGIS Pro, are executed more efficiently using Python scripting

*Step 4*: Finally, the XYZ files were converted into shapefiles using the "ASCII to Feature" tool in ArcGIS Pro, as shown in Fig. 4. Though this step still utilizes ArcGIS Pro, the preceding Python-based processing significantly reduces the dataset size and computational burden, making this final step faster and more efficient than processing raw LAZ/LAS files directly in ArcGIS Pro

By automating critical processing steps, this Python-based method minimizes manual intervention, reduces processing time, and allows for seamless handling of large datasets, making it a more scalable and practical solution for LiDAR-based surface-elevation-change analysis. However, a key limitation of the Python-based approach was the frequent updates to LiDAR processing libraries, such as pylas, laspy, and pcl, requiring some rebuilding processes. These updates increase processing time and pose challenges for research projects with strict timelines. (used in the Python-based method)

### 3.4 Matching and merging point features with different spatial coverage

The second phase of the Python-based workflow (Fig. 3) describes the preparation and matching of point features from the 2006 and 2017 LiDAR datasets to compute elevation differences. The workflow consists of six steps:

*Step 1*: Matching Spatial Coverage

The 2006 LiDAR dataset covers a larger geographic extent than the 2017 dataset, which is distributed across four files. To establish a common analysis area, the 2006 point feature classes were partitioned into four corresponding subsets matching the spatial extent of the 2017 datasets. Polygon boundaries representing the 2017 coverage were generated using the *Aggregate Points* tool in ArcGIS Pro (Fig. 1) and used to define the clipping extents.

*Step 2*: Clipping the 2006 Point Features

The 2006 point features were clipped to the spatial extent of each 2017 dataset using ArcGIS Pro ModelBuilder (Fig. 5a). This process produced four clipped feature classes corresponding to the four 2017 datasets while preserving the original feature attributes.

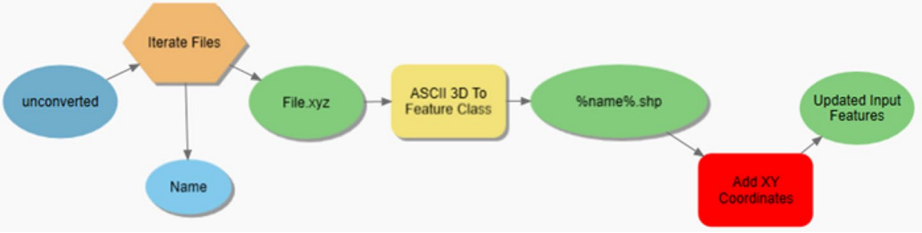
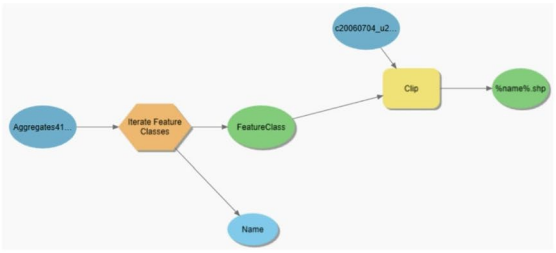



**Fig. 4** ArcGIS Pro model developed to convert XYZ to shapefile and add XY coordinates to shapefiles

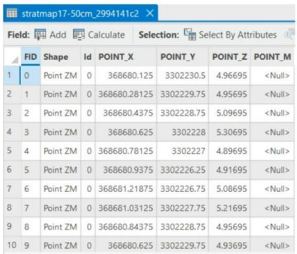
*Step 3*: Adding Coordinate and Elevation Attributes
Spatial coordinates (X and Y) and elevation values (Z) were added to the attribute table of each point feature using the *Add XY Coordinates* tool in ArcGIS Pro (Fig. 5b), providing the spatial attributes required for subsequent feature matching.
*Step 4*: Matching Point Features by Proximity
The *Near* tool in ArcGIS Pro was used to identify the nearest point feature in the 2017 dataset for each point feature in the 2006 dataset using a 1-m search distance. For each matched feature pair, the nearest feature identifier (*NEAR_FID*), separation distance (*NEAR_DIST*), and associated spatial information were appended to the attribute table of the 2006 feature class. This nearest-neighbor matching established corresponding point features between the two acquisition periods.
*Step 5*: Joining Elevation Attributes
To avoid duplicate field names, the elevation field (*POINT_Z*) in the 2017 feature class was renamed *POINT_Z_1*. A Python-automated *Join Field* operation was then performed using *NEAR_FID* and *FID* as the matching fields, transferring the 2017 elevation values to the corresponding 2006 point features.
*Step 6*: Calculating Elevation Differences
Elevation differences were computed for each matched feature pair using the *Calculate Field* tool within ArcGIS Pro. The elevation change (*Z_DIFFERENCE*) was calculated by subtracting the 2017 elevation (*POINT_Z_1*) from the corresponding 2006 elevation (*POINT_Z*). The resulting attribute table (Fig. 6) contains the coordinates and elevation of the 2006 features (*POINT_X*, *POINT_Y*, *POINT_Z*), the identifier and separation distance of the matched 2017 features (*NEAR_FID* and *NEAR_DIST*), the 2017 elevation (*POINT_Z_1*), and the computed elevation difference (*Z_DIFFERENCE*). Positive *Z_DIFFERENCE* values indicate surface elevation loss (potential subsidence), whereas negative values indicate surface elevation gain (potential uplift).

### 3.5 Focusing on areas along pipelines

This study focuses on pipeline corridors to identify areas of significant surface elevation change that may warrant further investigation for potential ground movement and its implications for pipeline integrity. To achieve this, a 100-ft wide buffer was created around the




a. Iteration model built in ArcGIS Pro for clipping the data from the first shapefiles based on the predefined polygons



stratmap17-50cm_2994141c2

Field: Add Calculate Selection: Select By Attributes

| | FID | Shape | Id | POINT_X | POINT_Y | POINT_Z | POINT_M |
|---|---|---|---|---|---|---|---|
| 1 | 0 | Point ZM | 0 | 368680.125 | 3302230.5 | 4.96695 | <Null> |
| 2 | 1 | Point ZM | 0 | 368680.28125 | 3302229.75 | 4.95695 | <Null> |
| 3 | 2 | Point ZM | 0 | 368680.4375 | 3302228.75 | 5.09695 | <Null> |
| 4 | 3 | Point ZM | 0 | 368680.625 | 3302228 | 5.30695 | <Null> |
| 5 | 4 | Point ZM | 0 | 368680.78125 | 3302227 | 4.89695 | <Null> |
| 6 | 5 | Point ZM | 0 | 368680.9375 | 3302226.25 | 4.91695 | <Null> |
| 7 | 6 | Point ZM | 0 | 368681.21875 | 3302226.75 | 5.08695 | <Null> |
| 8 | 7 | Point ZM | 0 | 368681.03125 | 3302227.75 | 5.21695 | <Null> |
| 9 | 8 | Point ZM | 0 | 368680.84375 | 3302228.75 | 4.95695 | <Null> |
| 10 | 9 | Point ZM | 0 | 368680.625 | 3302229.75 | 4.93695 | <Null> |

b. Attribute table of clipped shapefiles

**Fig. 5** Merging shapefile with different spacing resolution

| FID | Shape | Id | POINT_X | POINT_Y | POINT_Z | POINT_M | NEAR_FID | NEAR_DIST | NEAR_X | NEAR_Y | POINT_Z_1 | Z_Deferenc |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | Point ZM | 0 | 373248.598 | 3303927.164 | 10.53 | <Null> | 23316571 | 0.155247 | 373248.46875 | 3303927.25 | 11.32874 | -0.79874 |
| 1 | Point ZM | 0 | 373303.87 | 3303926.534 | 5.01 | <Null> | 23318522 | 0.164796 | 373304.03125 | 3303926.5 | 8.4087 | -3.3987 |
| 2 | Point ZM | 0 | 373295.032 | 3303927.058 | 8.4 | <Null> | 23319009 | 0.137121 | 373295.15625 | 3303927 | 4.16873 | 4.23127 |
| 3 | Point ZM | 0 | 373295.771 | 3303927.261 | 5.17 | <Null> | 23318347 | 0.023707 | 373295.75 | 3303927.25 | 13.98871 | -8.81871 |
| 4 | Point ZM | 0 | 373251.615 | 3303933.26 | 4.6 | <Null> | 23316092 | 0.014142 | 373251.625 | 3303933.25 | 10.76873 | -6.16873 |
| 5 | Point ZM | 0 | 373252.348 | 3303932.829 | 10.02 | <Null> | 23316107 | 0.08661 | 373252.3125 | 3303932.75 | 12.96872 | -2.94872 |
| 6 | Point ZM | 0 | 373278.869 | 3303931.681 | 5.62 | <Null> | 23317321 | 0.073475 | 373278.84375 | 3303931.75 | 15.33872 | -9.71872 |
| 7 | Point ZM | 0 | 373254.578 | 3303934.495 | 4.95 | <Null> | 23316774 | 0.016286 | 373254.5625 | 3303934.5 | 7.65871 | -2.70871 |
| 8 | Point ZM | 0 | 373250.156 | 3303934.546 | 4.81 | <Null> | 23316050 | 0.077805 | 373250.21875 | 3303934.5 | 12.88873 | -8.07873 |
| 9 | Point ZM | 0 | 373250.156 | 3303934.546 | 13.01 | <Null> | 23316050 | 0.077805 | 373250.21875 | 3303934.5 | 12.88873 | 0.12127 |
| 10 | Point ZM | 0 | 373374.656 | 3303928.898 | 4.87 | <Null> | 23324647 | 0.138708 | 373374.75 | 3303929 | 11.8887 | -7.0187 |
| 11 | Point ZM | 0 | 373448.322 | 3303925.311 | 4.74 | <Null> | 24520676 | 0.21796 | 373448.53125 | 3303925.25 | 12.27889 | -7.53889 |

**Fig. 6** Table generated by merging two shapefile sets transformed from two LiDAR datasets

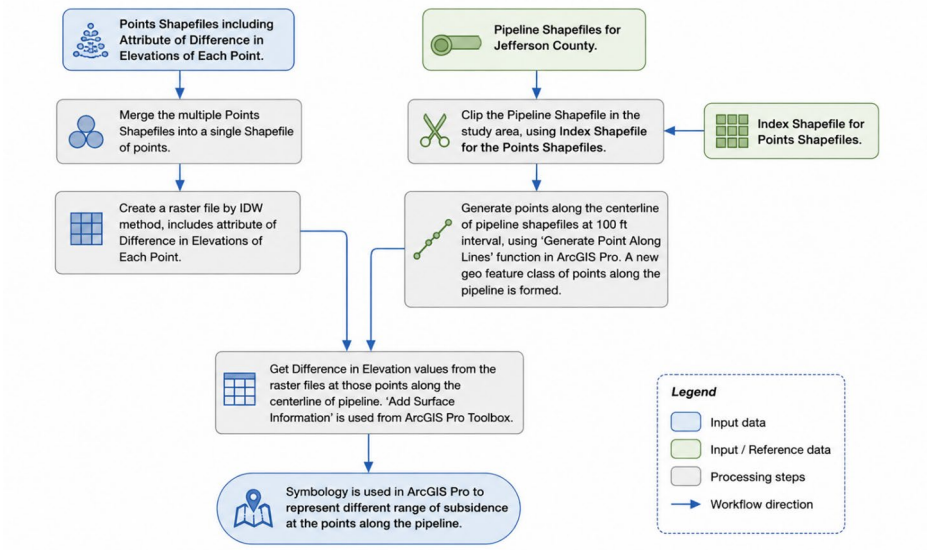



**Fig. 7** Workflow for generating and visualizing elevation difference (subsidence) along pipeline centerlines in Jefferson County, Texas

pipelines using elevation difference and pipeline shapefiles. Points within this buffer were extracted from the merged LiDAR dataset, as illustrated in Fig. 7, and processed in ArcGIS Pro.

This workflow integrates and compares the shapefiles to calculate surface elevation changes along pipeline corridors (see Sect. 4). By mapping these changes, the method identifies areas of notable elevation loss or gain that may warrant further investigation for potential ground movement and its implications for pipeline integrity. Its ability to process large datasets efficiently also provides a scalable approach for screening and monitoring surface elevation changes to support infrastructure assessment and risk mitigation.

## 4 Results

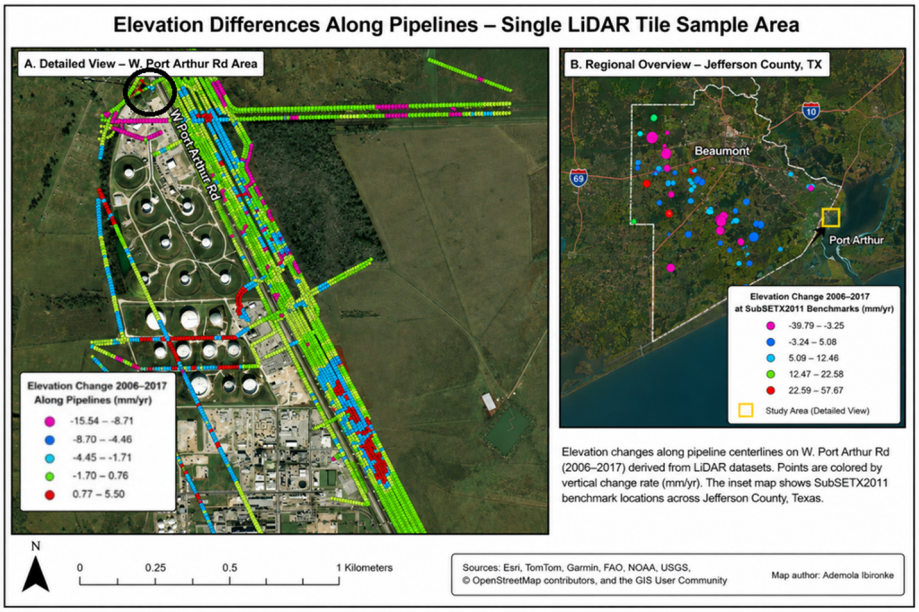
### 4.1 Smaller sample area results using ArcGIS pro-based workflow

Surface elevation changes between the 2006 and 2017 LiDAR datasets were calculated for each spatially matched point using the methods described in Sect 3. Figure 8a shows the spatial distribution of the classified elevation changes along the pipeline corridors within the smaller sample area, following the elevation change calculation. Most points were classified within the lower-magnitude elevation-change categories (green), while points with larger elevation-loss rates of 22.5–57.6 mm/year (red) occurred in localized clusters, including areas near the chemical plant silos and within approximately 50–100 m of West Port Arthur Road.

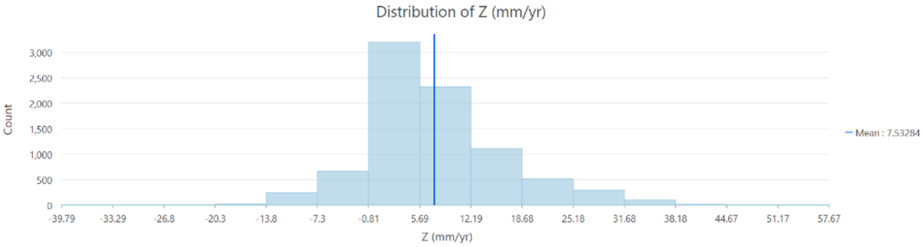
The histogram of elevation-change rates, divided into 15 bins (Fig. 8b), shows that the dominant range was −0.81 to 5.69 mm/year, with an average elevation-change rate of 7.5 mm/year. The distribution was asymmetric and skewed toward positive values, representing surface elevation loss under the sign convention used in this study. The maximum observed elevation-loss rate was 57.67 mm/year, while the maximum elevation-gain rate was 39.79 mm/year.

### 4.2 Larger study area results using the Python-enhanced workflow

The Python-enhanced workflow was applied to the full study area to demonstrate its capability for efficient large-scale processing of multi-temporal LiDAR datasets. Expanding the analysis beyond the representative sample area enabled the identification and visualization of surface elevation changes across a broader spatial extent (Fig. 9a). The analysis revealed a wider range of elevation changes than observed in the smaller study area (as reported in Fig. 8a). Then, the calculated values were classified into five categories using the Jenks Natural Breaks (JNB) method (Jenks, 1967): larger elevation gain, elevation gain, small elevation change (±), elevation loss and larger elevation loss (as reported in Fig. 9b). The JNB is an iterative technique that groups data into clusters, minimizing variance within each class by reallocating data points to optimize classification. This method was widely used in the studies such as land use analysis (Li and Xu, 2020), deep learning-based location estimation (Hou et al., 2022), and thematic map creation (Abboud et al., 2020). Elevation change was calculated as the 2006 elevation minus the corresponding 2017 elevation. Therefore, positive values indicate surface elevation. Application of the Jenks Natural Breaks (JNB) classification identified a dominant elevation-change range of 0.62 to 1.08 m (56.4–98 mm/year) over the 11-year period from 2006 to 2017. Figure 10 shows cumulative LiDAR-derived elevation differences from 2006 to 2017, with most matched points concentrated in the broad central interval and relatively few at the extremes. The mean difference was 0.2568 m over 11 years (23.35 mm/year), with extremes of 11.92 m and − 14.4 m. The distribution was skewed toward positive elevation differences. Despite differences in spatial extent and processing methods, comparisons at overlapping locations showed broadly comparable spatial patterns between the ArcGIS Pro- and Python-derived results, as discussed further in Sect. 5.3.




(a) Elevation changes along pipelines on W. Port Arthur Rd (Single Tile Area) with benchmark (blue dot in black circle)

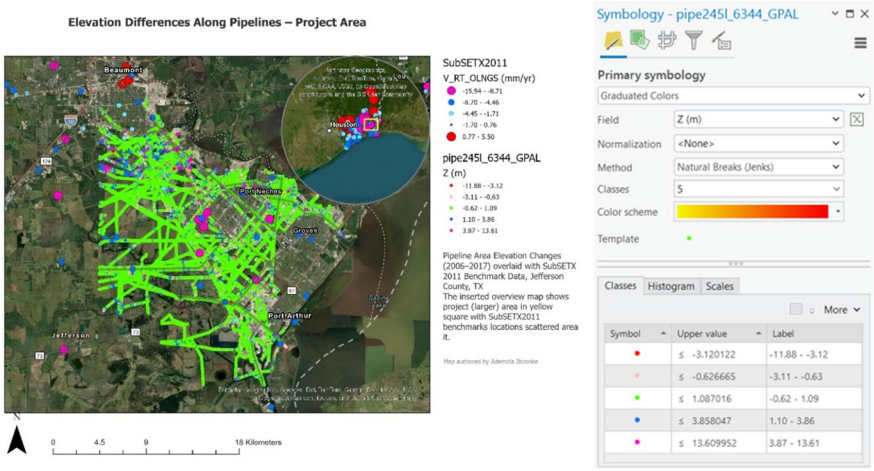



(b) Histogram of elevation changes in the area with pipelines along W. Port Arthur Rd, Beaumont, Texas

**Fig. 8** Elevation change values of points along pipelines in smaller sample area (see Table 1) via Method 1. **a** Elevation changes along pipelines on W. Port Arthur Rd. with benchmark (in the black circle on the left panel) and critical elevation loss areas (red dots on the left panel), **b** Histogram of elevation changes in the area with pipelines along W. Port Arthur Rd., Beaumont, Texas.

### 4.3 Comparison of Method 1 (ArcGIS Pro-based) and Method 2 (Python-enhanced)

Before comparing the two workflows, it is important to note that they were applied to different spatial extents for practical and computational reasons. The ArcGIS Pro-based workflow, which requires substantial manual intervention, was demonstrated using a representative single LiDAR tile because its processing time increases considerably as additional tiles are incorporated. In contrast, the Python-enhanced workflow was specifically developed to automate batch processing and was therefore evaluated over the entire project area comprising multiple LiDAR tiles to demonstrate its capability for efficient large-scale analysis. Although the workflows were applied to different spatial extents, comparisons were performed using overlapping locations and representative point features to evaluate the consistency of elevation-change measurements while highlighting differences in computational




(a) Pipeline Area Elevation Changes (2006–2017) with SubSETX 2011 Benchmark Data, Jefferson County, TX

(b) Results Classification based on Jenks Natural Breaks

**Fig. 9** Elevation differences along pipelines in larger area. **a** Pipeline area elevation changes (2006–2017) with SubSETX 2011 Benchmark Data, Jefferson County, Texas, **b** Results classification based on Jenks Natural Breaks (JNB)

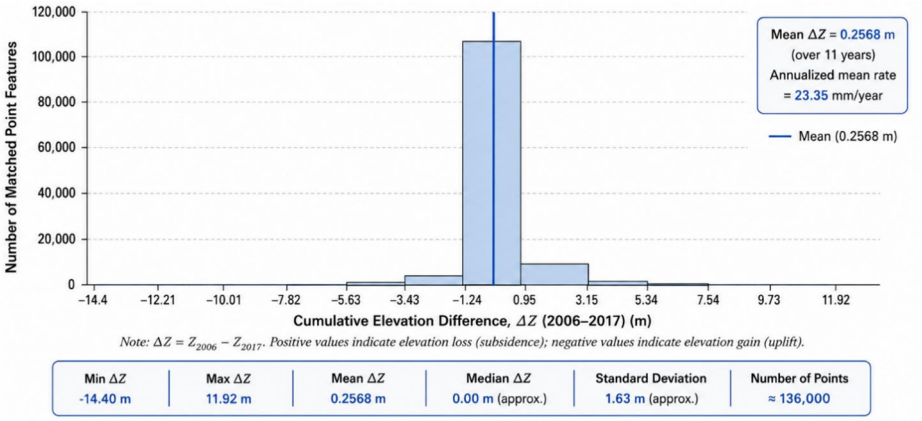



**Fig. 10** Distribution of cumulative elevation differences derived from LiDAR datasets between 2006 and 2017

efficiency and scalability. Accordingly, the comparison emphasizes workflow performance, scalability, and the consistency of observed spatial patterns of elevation change rather than a direct one-to-one comparison of results derived from identical datasets.

### 4.3.1 Processing efficiency and time comparison

Figure 11 presents the measured processing times for the ArcGIS Pro-based and Python-enhanced workflows at the single-tile and project scales. Individual LiDAR tiles ranged

from approximately 0.3 to 0.5 GB, with typical processing tasks involving 1–2 GB of input data.

For the representative single LiDAR tile processed using both workflows, the ArcGIS Pro-based workflow required approximately 3.5 h, whereas the Python-enhanced workflow required approximately 0.8 h, corresponding to an observed processing-time reduction of approximately 77% (Fig. 11). Because the ArcGIS Pro workflow was not executed over the complete multi-tile project area under the same conditions, no direct project-scale processing-time comparison was made. The Python-enhanced workflow was subsequently applied to the larger project dataset, where its batch-processing capability enabled automated processing of multiple LiDAR tiles. Accordingly, the single-tile results provide the direct processing-time comparison between the workflows, whereas the larger-area application demonstrates the scalability of the automated workflow.

### 4.3.2 Processing efficiency and time comparison

Across overlapping zones, the ArcGIS Pro-based and Python-enhanced workflows produced broadly comparable elevation-change results, although differences in magnitude were observed at individual points and within localized areas. Numerical comparisons at selected overlapping locations showed deviations between the elevation-change values generated by the two workflows, with larger discrepancies observed in some areas of pronounced elevation change. Despite these point-scale deviations, both workflows identified similar locations and general distributions of surface elevation loss and gain across the overlapping areas. Both workflows produced similar spatial patterns of elevation change. The numerical deviations and spatial similarities are observed further examined in Sect. 4.4.

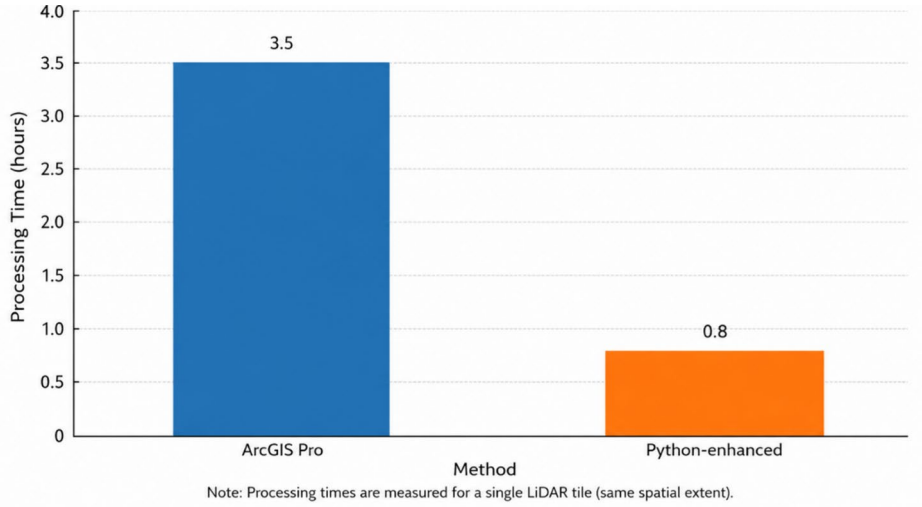



**Fig. 11** Processing time comparison between Method 1 (ArcGIS Pro-based) and Method 2 (Python-enhanced)

### 4.4 Comparison of LiDAR-derived elevation changes with GPS-based height modernization benchmarks and InSAR observations

LiDAR-derived surface-elevation changes were compared with the SubSETX2011 benchmark survey (Kruger 2019), which reported rates ranging from approximately −15.54 mm/year (subsidence) to +3 mm/year (uplift). The LiDAR analysis used the opposite sign convention, with positive values indicating elevation loss. Figure 12 overlays the ArcGIS Pro-based and Python-enhanced results with SubSETX2011 benchmarks near the Beaumont pipeline corridors. Within the West Port Arthur Road study area, the V_RT_OLNGS benchmark recorded −4.1 mm/year, while nearby LiDAR-derived elevation-loss rates were 3.8 and 1.6 mm/year for ArcGIS Pro Points A and C and 2.72 mm/year for Python-enhanced Point B (Fig. 13). Thus, the nearest LiDAR-derived values ranged from 1.6 to 3.8 mm/year compared with the benchmark value of −4.1 mm/year, accounting for the opposite sign conventions.

Additional numerical differences between the ArcGIS Pro-based and Python-enhanced workflows were observed at selected locations near the chemical plant along West Port Arthur Road (Fig. 14). At one location, the ArcGIS Pro workflow produced a surface elevation-loss rate of 28.21 mm/year, whereas the nearest Python-derived point produced 0.056 m over 11 years, equivalent to 5.09 mm/year (Fig. 14a, b). At another location, the ArcGIS Pro-derived rate was 25 mm/year, compared with 0.07 m over 11 years (6.4 mm/year) from the Python-enhanced workflow (Fig. 14c, d). The compared points in the latter pair were approximately 2 m apart. Differences were therefore observed between the ArcGIS Pro- and Python-enhanced measurements at these locations.

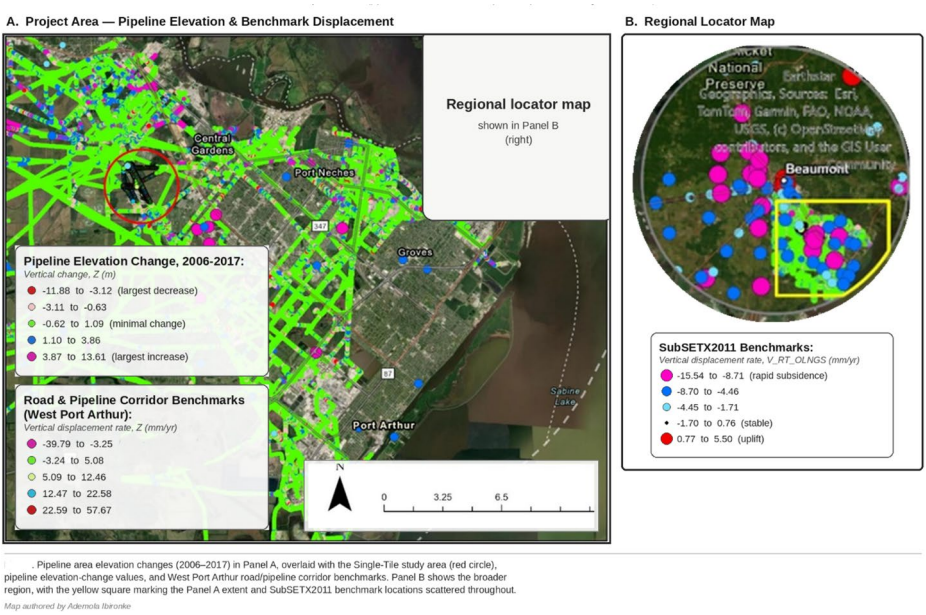



**Fig. 12** Overlay of ArcGIS Pro-based (in red circles), Python-enhanced, and SubSTEX survey results near pipelines in Beaumont, Texas




(a) SubSETX2011 benchmark (blue dot), elevation change rate of – 4.1mm/yr

(b) ArcGIS Pro point A nearest to benchmark, elevation change rate of 3.8 mm/yr

(c) Python calculated point B, elevation change rate of 0.03 m over 11 years (2.72mm/yr.)

(d) ArcGIS Pro point C, elevation change rate of 1.6 mm/yr

**Fig. 13** Elevation changes measurement at survey benchmarks and nearest points along pipelines. **a** SubSETX2011 benchmark (blue dots) with elevation change rate of – 4.1 mm/yr, **b** ArcGIS Pro point A nearest to benchmark with elevation change rate of 3.8 mm/yr, **c** Python calculated point B with elevation change rate of 0.03 m over 11 years (2.72 mm/yr), **d** ArcGIS Pro point C with elevation change rate of 1.6 mm/yr

The nearest SubSETX benchmark to the locations shown in Fig. 14 was approximately 400 m away. Consequently, no benchmark measurement was available at the same or immediately adjacent locations for direct numerical comparison with the LiDAR-derived values shown in Fig. 14.

For comparison with an additional dataset, InSAR measurements reported by Shrestha (2024) for the same general study area were examined. The analysis used Sentinel-1 scenes acquired between 2016 and 2022, with the December 15, 2016 scene serving as the reference. The reported InSAR-derived vertical displacement rates varied annually and generally remained within ±2–3 cm/year (±20–30 mm/year). RTK/GNSS SubSETX measurements reported rates ranging from −5.5 mm/year (uplift) to 15.54 mm/year (subsidence) in 2011 and from −10 mm/year (uplift) to 13.4 mm/year (subsidence) in 2023 (Owoyombo, 2023).




(a) ArcGIS Pro elevation change rate of 28.21mm/yr at a potentially critical point near chemical plant

(b) Python calculated elevation change rate of 0.056m over 11 yrs (5.09 mm/yr) of nearest point to (a)

(c) ArcGIS Pro elevation change rate of 25mm/yr at another potentially critical point

(d) Python calculated elevation change rate of 0.07 m over 11 years (6.4 mm/yr.) of same point as (c)

**Fig. 14** Elevation Changes at Potentially Critical Points near a Chemical Plant on West Port Arthur Road. **a** ArcGIS Pro elevation change rate of 28.21 mm/yr at a potentially critical point near a chemical plant, **b** Python calculated elevation change rate of 0.056 m over 11 years, (5.09 mm/yr) of nearest point to (a), **c** ArcGIS Pro elevation change rate of 25 mm/yr at another potentially critical point, **d** Python calculated elevation change rate of 0.07 m over 11 years (6.4 mm/yr.) of same point as (c)

The LiDAR analysis produced an average surface elevation-change rate of approximately 23.35 mm/year. The LiDAR, InSAR, and RTK/GNSS measurements represent different acquisition periods and were therefore reported as separate numerical observations rather than directly equivalent measurements.

## 5 Discussion

This study evaluated two workflows for detecting surface-elevation changes along pipeline corridors from multi-temporal LiDAR acquired in 2006 and 2017. The results demonstrate both the potential and the limitations of using repeated LiDAR surveys for large-area screening of surface change. The observed elevation differences were spatially variable and

included both elevation loss and elevation gain, while the Python-enhanced workflow substantially reduced the processing effort required for larger datasets. However, the detected changes should not be interpreted automatically as subsidence or uplift because multi-temporal surface-elevation differences can also reflect land development, erosion or deposition, vegetation and classification effects, differences in point sampling, and other changes occurring between acquisition dates. Accordingly, the discussion focuses on the spatial patterns of measured elevation change, workflow performance, cross-comparison among available measurements, sources of discrepancy, and the potential role of the approach as a screening tool for infrastructure monitoring.

### 5.1 Spatial distribution of elevation changes

Both the smaller ArcGIS Pro sample and the larger Python-processed dataset showed spatially varied surface-elevation changes along the pipeline corridors. Within the smaller sample area along West Port Arthur Road, the distribution was asymmetric toward elevation loss, with a dominant range of −0.81 to 5.69 mm/year and an average elevation-change rate of 7.5 mm/year (Fig. 9b). Larger positive differences were concentrated at selected locations near the chemical plant and West Port Arthur Road, while much of the mapped corridor exhibited comparatively smaller changes. These localized concentrations indicate that surface change was not spatially uniform across the corridor. The larger project area analysis similarly identified a broad range of positive and negative elevation differences. The histogram produced from the Python-enhanced workflow ranged predominantly from −1.02 to 0.96 m over the 11-year interval, with an average elevation loss of 0.25 m (23.35 mm/year). The distribution was also asymmetric toward elevation loss. The differences between the distributions observed in the smaller and larger study areas also illustrate the influence of spatial extent and sampling on summary statistics; results obtained from a single representative tile should therefore not be assumed to characterize the complete project area.

Importantly, these patterns represent surface-elevation change rather than confirmed ground subsidence. The LiDAR comparison determines whether the measured ground-surface elevation at spatially corresponding locations changed between 2006 and 2017, but it does not independently identify the physical process responsible for that change. For example, areas near roads and industrial facilities may have experienced grading, excavation, filling, construction, erosion, sediment deposition, or other surface modifications during the 11-year interval. Consequently, locations exhibiting comparatively large elevation loss are described as surface-change locations requiring further investigation rather than subsidence hotspots.

This distinction modifies the interpretation of Figs. 9, 8 and 10 without reducing their usefulness. The spatial clustering of larger changes can serve as an initial screening layer for identifying locations where additional information—including land-use history, more frequent elevation observations, InSAR time series, or field surveys—may be warranted. The contribution of the LiDAR analysis is therefore the systematic detection and mapping of where substantial surface changes occurred, rather than determination of their cause.

### 5.2 Performance of the python-enhanced workflow

The clearest advantage of the Python-enhanced workflow was computational efficiency. The ArcGIS Pro workflow required approximately 3.5 h to process the representative single tile, compared with approximately 0.8 h using the Python-enhanced workflow. At the project scale, the Python-enhanced workflow was subsequently applied to the larger multi-tile project area to evaluate its batch-processing capability. Figure 11 therefore demonstrates that the benefit of the Python-Enhanced Workflow became particularly important as the number of LiDAR files increased. The original workflow involves multiple filtering, conversion, aggregation, and spatial-join operations.

The improvement is primarily attributable to workflow automation rather than evidence that the Python approach produces inherently more accurate elevation measurements. Automating repetitive file conversions, point-cloud operations, spatial matching, and attribute calculations reduces repeated user interaction and allows multiple datasets to be processed systematically. This distinction is important because processing efficiency and measurement accuracy are separate performance criteria. The substantial reduction in runtime can be supported directly by the measured processing times, whereas accuracy cannot be inferred from processing speed.

The point-level comparisons show that the Python-enhanced and ArcGIS Pro workflows do not produce numerically equivalent results. At selected locations shown in Fig. 14, ArcGIS Pro produced elevation-loss rates of 28.21 and 25 mm/year, whereas nearby Python-derived values were 5.09 and 6.4 mm/year, respectively. These differences are substantially greater than the 2–3 mm/year deviations observed at some other overlapping locations. Therefore, a single generalized error range should not be assigned to the two workflows based on the available comparisons.

Rather, the results indicate that the workflows can produce similar regional spatial patterns while differing substantially at individual locations. The Python-enhanced workflow consequently appears primarily as a more scalable implementation for processing large multi-temporal LiDAR datasets. Its principal contribution demonstrated in this study is the reduction of processing time and manual intervention while retaining the ability to map spatial patterns of elevation change over a much larger area.

### 5.3 Agreement between ArcGIS Pro, Python, benchmark, and InSAR measurements

Comparison among the different measurement sources requires particular caution because they differ in spatial resolution, temporal coverage, measurement technique, and observation location. Near the V_RT_OLNGS SubSETX2011 benchmark, the benchmark reported −4.1 mm/year using its convention of negative values for subsidence. Nearby LiDAR-derived surface-elevation-loss rates were 3.8 and 1.6 mm/year from ArcGIS Pro and 2.72 mm/year from the Python-enhanced workflow. After accounting for the opposite sign convention, these measurements indicate the same direction of surface change and are of similar magnitude. This provides a useful local cross-check, but one location is insufficient to establish statistical agreement or validate either LiDAR workflow. The comparisons in Fig. 14 reinforce this cautious interpretation. Although both LiDAR workflows identified elevation loss at the selected locations, their estimated magnitudes differed considerably. For example, 28.21 mm/year from ArcGIS Pro was compared with 5.09 mm/year from

the nearest Python-derived point, while another comparison yielded 25 and 6.4 mm/year, respectively. Thus, agreement in the direction or spatial occurrence of elevation change should not be interpreted as point-by-point numerical agreement.

The InSAR and RTK/GNSS observations provide additional context rather than independent validation. InSAR observations for 2016–2022 generally fell within ± 2–3 cm/year, while SubSETX surveys reported deformation rates ranging from − 5.5 mm/year for uplift to 15.54 mm/year for subsidence in 2011 and from − 10 mm/year for uplift to 13.4 mm/year for subsidence in 2023. The LiDAR-derived average elevation-change rate of approximately 23.35 mm/year is therefore within the same broad order of magnitude as some independently reported deformation measurements, but this similarity does not establish correlation.

Most importantly, annualizing each dataset to mm/year does not make the observations temporally equivalent. The LiDAR analysis represents net change between 2006 and 2017, the InSAR observations cover 2016–2022, and the benchmark measurements represent other observation periods. Surface deformation may vary considerably through time, particularly in areas affected by groundwater withdrawal, industrial activity, climatic variability, or land-surface modification. Consequently, the three datasets should be treated as independent observations providing qualitative context, not as simultaneous measurements suitable for formal accuracy assessment. Quantitative validation would require spatially coincident measurements collected over substantially overlapping time intervals.

### 5.4 Sources of discrepancy and study limitations

Several factors can explain the differences observed among the ArcGIS Pro, Python-enhanced, benchmark, and InSAR results. First, the LiDAR workflows rely on spatially matching observations from two-point clouds acquired 11 years apart. The exact ground locations sampled during the two flights are not necessarily identical. Nearest-neighbor matching therefore associates spatially close observations rather than repeated measurements of precisely the same physical point. On sloping or irregular terrain, even a small horizontal displacement between matched points can produce an apparent vertical difference unrelated to actual ground deformation.

Second, differences between the ArcGIS Pro and Python-enhanced workflows can arise from their point-processing and spatial-matching procedures. Variations in point density, spatial coverage, nearest-neighbor selection, filtering, and intermediate data transformations can produce different point pairs and therefore different elevation-change values. The substantially different values observed at the Fig. 14 locations demonstrate that these processing effects can be important at the individual-point scale. This is particularly relevant where nearby points occur on surfaces with strong local elevation gradients.

Third, the availability of ground-based reference measurements was limited. Within the smaller West Port Arthur Road study area, only one SubSETX benchmark was sufficiently close for the comparison presented in Fig. 13, while the nearest benchmark to Fig. 14 locations was approximately 400 m away. The benchmark network therefore does not provide the spatial density necessary for systematic point-by-point validation of LiDAR-derived changes throughout the pipeline corridor.

Temporal mismatch presents an additional limitation. The LiDAR, InSAR, 2011 benchmark, and 2023 RTK/GNSS observations do not represent identical measurement periods.

Annualizing their changes facilitates comparison of magnitude but cannot account for temporal variability in surface movement. Similarly, Sentinel-1 InSAR and airborne LiDAR measure the surface using fundamentally different sensing geometries, spatial resolutions, and processing procedures. Their reported displacement values therefore should not be expected to coincide exactly. A more fundamental limitation concerns attribution. Surface elevation loss detected between two LiDAR acquisitions cannot by itself be attributed to subsidence. Land development, grading, excavation, erosion, deposition, vegetation-related classification differences, changes to roads or industrial facilities, and other surface disturbances may produce similar elevation differences. The current analysis does not contain sufficient temporal or ancillary information to distinguish systematically among these mechanisms. Accordingly, the study identifies surface-elevation changes near pipelines, not confirmed subsidence events or pipeline-failure risk.

Future validation should therefore incorporate more spatially and temporally coincident observations, including repeated GNSS measurements, permanent or temporary survey control points, higher-frequency LiDAR acquisitions, and InSAR time series covering the same observation period. Land-use and construction records could additionally help distinguish gradual ground deformation from anthropogenic surface modification. These additions would allow the screening workflow developed here to progress toward quantitative deformation monitoring.

### 5.5 Practical implications for large-scale subsidence monitoring

Despite these limitations, the workflow has practical value as a large-area screening approach for infrastructure monitoring. Transportation, pipeline, and other infrastructure networks can extend across areas too large for continuous ground-based surveying. Multitemporal LiDAR provides a means of systematically examining previously acquired elevation datasets and identifying locations where surface elevations have changed substantially between surveys. The Python-enhanced workflow is particularly useful in this context because its measured processing-time reductions demonstrate that large multi-tile datasets can be evaluated with substantially less processing effort than the predominantly ArcGIS Pro-based workflow. The appropriate role of the method is therefore not to independently diagnose subsidence or predict pipeline failure, but to support a tiered monitoring strategy. LiDAR-derived surface-change maps can first identify locations exhibiting unusually large or spatially concentrated elevation changes. These locations can then be prioritized for closer examination using higher-frequency InSAR, GNSS or conventional surveys, site inspections, land-use records, and other geotechnical or infrastructure information. In this framework, LiDAR serves as a broad screening layer, while more direct and temporally resolved measurements are used to determine the cause and engineering significance of the detected changes.

For pipeline corridors, this distinction is particularly important. A large elevation difference near a pipeline does not necessarily imply that the pipeline has experienced deformation or is at imminent risk. It indicates instead that the surrounding land surface changed sufficiently between acquisition periods to warrant additional investigation. Combining such observations with pipeline characteristics, soil conditions, construction history, groundwater information, and repeated deformation measurements would be necessary before making conclusions regarding infrastructure risk.

The principal methodological contribution of this study is therefore the demonstration of a scalable workflow for detecting and prioritizing surface-elevation changes across large infrastructure corridors. The substantial reduction in processing time achieved by the Python-enhanced implementation supports its use for regional screening and repeated analysis as new LiDAR datasets become available. With improved temporal coverage and independent field measurements, the framework could form one component of a broader multi-sensor monitoring system for identifying areas where ground deformation may warrant detailed engineering assessment.

## 6 Conclusions

This study developed and compared two workflows for processing multi-temporal LiDAR data to quantify surface elevation changes along crude oil pipeline corridors in Southeast Texas: a conventional ArcGIS Pro-based workflow and a Python-enhanced GIS workflow designed to automate data processing. Both workflows successfully extracted LiDAR-derived elevation information, generated point feature datasets, and computed elevation differences that can be used to identify areas of potential ground movement affecting pipeline infrastructure. The Python-enhanced workflow demonstrated automation, improved scalability, and computational efficiency, making it well suited for processing large LiDAR datasets while producing spatial patterns of elevation change comparable to those obtained using the ArcGIS Pro-based approach.

Analysis of LiDAR data acquired between 2006 and 2017 identified several localized hotspots exhibiting substantial surface elevation changes that may warrant further monitoring because ground deformation, whether caused by subsidence, uplift, erosion, or land development, can adversely affect pipeline integrity. Comparisons with differential Interferometric Synthetic Aperture Radar (InSAR) observations and the available SubSETX benchmark provided qualitative support for the observed elevation-change patterns. However, because only one benchmark was located sufficiently close to the analyzed LiDAR point features and the datasets represent different temporal periods, these comparisons should be interpreted as qualitative corroboration rather than comprehensive quantitative validation. Accordingly, the study demonstrates the potential of multi-temporal LiDAR for identifying areas of surface elevation change associated with possible subsidence rather than establishing definitive subsidence rates.

The findings also highlight the importance of considering other geomorphological processes, including land development and erosion, alongside subsidence and uplift, because each can influence pipeline stability and contribute to infrastructure risk. While the proposed methodology shows promise as a rapid and scalable approach for infrastructure monitoring, future studies incorporating a denser network of ground-based benchmarks, continuous GNSS observations, and additional multi-temporal datasets are needed to strengthen quantitative validation and improve the reliability of subsidence assessments. Such efforts will further enhance the applicability of LiDAR-based monitoring for infrastructure management and risk mitigation.

For clarity, the principal strengths and limitations of the proposed methodology are summarized in Table 2.

Based on this study, the following are the possible direction of future studies.

**Table 2** Summary of Strength and Limitations

| Aspect | Strengths | Limitations |
|---|---|---|
| Flexibility in Data Processing | - Employed two methodologies: ArcGIS Pro tools and Python libraries.<br>- Python provided advanced data processing capabilities, enabling handling of larger datasets and more complex analyses. | - Discrepancies in elevation change calculations between ArcGIS Pro and Python workflows due to differences in resolution, filtering, or outlier handling. |
| Innovative Methodologies | - Introduced novel workflows for converting LiDAR data into shapefiles and calculating elevation changes.<br>- Used Python libraries (Laspy, Pulpy, Open3d) for advanced data processing. | - Frequent updates to Python libraries (e.g., Laspy, Pulpy) required new processing environments, increasing complexity and processing time. |
| Multi-Temporal Analysis | - Utilized LiDAR datasets from 2006 and 2017 to capture long-term elevation changes.<br>- Provided insights into subsidence patterns over an 11-year period. | - The 11-year gap between datasets limits the ability to detect short-term or seasonal variations in subsidence. |
| Integration with Other Data Sources | - Integrated LiDAR data with InSAR and ground-based benchmark surveys (SubSETX 2011).<br>- provides independent qualitative corroboration from different measurement techniques, enhancing reliability. | - Sparse distribution of ground survey benchmarks limited validation across the study area.<br>- Only one benchmark was available in the smaller sample area, reducing validation accuracy in localized regions. |
| Focus on Critical Infrastructure | - Focused on crude oil pipelines in Southeast Texas, a region with high energy infrastructure density.<br>- Identified hotspots with significant elevation changes, providing actionable insights for risk mitigation. | - Environmental factors (e.g., vegetation) affected LiDAR and InSAR data accuracy, particularly in heavily vegetated areas. |
| Data Accuracy and Resolution | - LiDAR provided high-resolution data for detailed elevation change analysis. | - Discrepancies in elevation change calculations between workflows due to differences in data resolution and filtering techniques. |
| Temporal Inconsistencies | - Multi-temporal analysis provided long-term trends in subsidence. | - The 11-year gap between datasets may miss rapid or seasonal elevation changes caused by geomorphological events or human activities. |
| Sparse Ground Survey Data | - Ground survey benchmarks provided some effective comparison for LiDAR results. | - Sparse distribution of benchmarks limited validation across the study area.<br>- Only one benchmark was available in the smaller sample area, reducing validation accuracy in localized regions. |
| Challenges in Data Processing | - Python-based workflow allowed for advanced data processing and handling of larger datasets. | - Frequent updates to Python libraries increased processing time and complexity.<br>- Smaller sample size in ArcGIS Pro workflow may have missed important variations in elevation changes. |
| Environmental and Vegetation Limitations | - Analyzed data during vegetation dormancy to mitigate InSAR limitations. | - InSAR struggles to penetrate dense vegetation, limiting its effectiveness in certain areas.<br>- Environmental factors (e.g., vegetation) could impact the accuracy of subsidence measurements. |
| Trade-offs Between Efficiency and Accuracy | - Smaller sample size in ArcGIS Pro workflow improved computational efficiency. | - Smaller sample size may have missed important variations in elevation changes, reducing the generalizability of results. |

*Increased Temporal Resolution*: Incorporating more frequent LiDAR data collection, such as annual or biennial surveys, would provide a more detailed understanding of short-term elevation changes and their impact on pipeline infrastructure

*Enhanced Ground Validation*: Expanding the network of ground-based benchmarks and integrating additional data sources, such as GPS and InSAR, would improve the accuracy and reliability of subsidence measurements

*Improved Data Processing Techniques*: Developing more robust and automated workflows for LiDAR data processing, particularly in Python, could reduce the complexity and time required for data analysis. Additionally, exploring machine learning techniques for filtering and analyzing LiDAR data could enhance the accuracy of subsidence predictions

*Broader Spatial Coverage*: Expanding the study area to include more regions with high pipeline density would provide a more comprehensive assessment of subsidence risks across Southeast Texas

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**Author contributions** Ademola Ibironke, Fayez Albalawi and Aawaz together developed the models. Fayez Albalawi developed the initial model in ArcGIS Pro. Ademola developed Python-based model and applied to the study on the area along pipelines in Southeast Texas. Aawaz helped in the case study. Ademola and Fayez drafted the initial draft of the paper. Ademola also revised the paper and responded to all reviewers' comments. Dr. Xing Wu led the team in this project. He proposed the initial idea to secure the research funds to support the students and provided the guidance to graduate students in research directions and data analysis. Based on the work of Ademola, he finalized the initial and revised drafts, as well as the response to reviewers. Dr. Joseph Kruger, as the Co-Principal Investigator of this project, provided on-site surveying data for data verification (with LiDAR data), and offered guidance on data verification and analysis, as well as data interpretation and data comparison.

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**Data availability** Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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